

Modern Physics

$$1a) \quad n=1 \quad \downarrow \\ V_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{m e^2} \Rightarrow r_1 = \frac{4\pi\epsilon_0 \hbar^2}{m e^2} = a_0$$

$$V_n = \frac{1}{n} \frac{\hbar}{m a_0} \Rightarrow V_1 = \frac{\hbar}{m a_0}$$

If we increase the atomic number of the atom, there will be an increase in the mass and increase in positive charge inside the nucleus, causing increase in attraction between the nucleus and electron. Therefore, the radius will become smaller and the speed of electron will increase. //

$$b) \textcircled{i} \quad E = hf \quad \lambda = \frac{c}{f} \Rightarrow E = \frac{hc}{\lambda} \therefore \Delta E = \frac{hc}{\lambda' - \lambda}$$

$$\lambda' - \lambda = \frac{h}{m_0 c} (1 - \cos\theta)$$

$$\lambda' - \lambda = \frac{6.63 \times 10^{-34} \text{ J}\cdot\text{s}}{(9.11 \times 10^{-31} \text{ kg}) \times (3 \times 10^8 \text{ m/s})} \times 2$$

$$\lambda' - \lambda = 4.86 \times 10^{-12} \text{ m}$$

$$\Delta E = \frac{6.63 \times 10^{-34} \text{ J}\cdot\text{s} \times (3 \times 10^8 \text{ m/s})}{4.86 \times 10^{-12} \text{ m}} = 4.09 \times 10^{-14} \text{ J} //$$

$$\textcircled{ii} \quad E = \frac{hc}{\lambda} \\ = \frac{(6.63 \times 10^{-34} \text{ J}\cdot\text{s}) \times (3 \times 10^8 \text{ m/s})}{500 \times 10^{-9} \text{ m}}$$

$$= 0.03978 \times 10^{-17} = 3.97 \times 10^{-19} \text{ J} //$$

c) (i) $R(T) = \epsilon \sigma T^4$ (Power per unit area and unit wavelength)

$$\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$$

$$\lambda_{\text{max}} T = 2.898 \times 10^{-3} \text{ m K}$$

Assuming that the average human body temperature is 37°C

$$T = 310 \text{ K}$$

$$\therefore \lambda_{\text{max}} = \frac{2.898 \times 10^{-3} \text{ m K}}{310 \text{ K}} = 9.35 \times 10^{-6} \text{ m}$$

⊥ (black body)

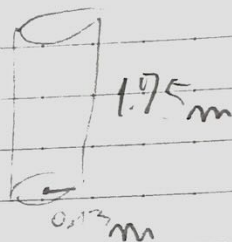
Power $W = \epsilon \sigma A T^4$

$$= (5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}) \times (310^4) \times [2\pi (0.13)(1.75)$$

$$+ 2\pi (0.13)^2]$$

$$= 884 \text{ W}$$

$$\rightarrow 884 \times 24 \times 60 \times 60 = 6.95 \times 10^7 \text{ J in a day}$$



(ii) $1 \text{ kcal} = 4184 \text{ J}$

$$E = 2000 \times 4184 = 8.34 \times 10^6 \text{ J}$$

The results do not make sense. That might be because we considered the human body to work as a black body.

2d) ①

$$b = \frac{z_1 z_2 e^2}{8\pi\epsilon_0 k} \cot \frac{\theta}{2}$$

$$z_1 = 2$$

$$z_2 = 79$$

$$b = \frac{2 \cdot 79 \times (1.6 \times 10^{-19} \text{ C})^2}{8 \times \pi (8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2 \times (7.7 \times 10^6 \text{ Nm})) \times 1.6 \times 10^{-19}} \cot \left(\frac{1^\circ}{2} \right)$$

$$= 1.69 \times 10^{-12} \text{ m}$$

②

$$b = \frac{2 \times 79 \times (1.6 \times 10^{-19})^2}{8 \pi (8.85 \times 10^{-12} \text{ C}^2/\text{Nm}^2 \times (7.7 \times 10^6 \text{ Nm})) \times (1.6 \times 10^{-19})} \cot \left(\frac{90^\circ}{2} \right)$$

$$= 1.47 \times 10^{-14} \text{ m}$$

b)

$$f = \pi n t \left(\frac{z_1 z_2 e^2}{8\pi\epsilon_0 k} \right)^2 \cos^2 \frac{\theta}{2}$$

$$\frac{f(2, \text{Au})}{f(2, \text{Al})} = \frac{(79)^2}{(13)^2} = 36.93 //$$

c)

$$L = mvr$$

$$r_1 = a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m e^2} = 0.53 \times 10^{-10} \text{ m}$$

$$L_1 = 9.11 \times 10^{-31} \text{ kg} \times (2.17 \times 10^6 \text{ m/s}) \times (0.53 \times 10^{-10} \text{ m})$$

$$= 1.05 \times 10^{-34} \text{ kg m}^2/\text{s}$$

$$V_1 = \frac{\hbar}{m r_1} = \frac{1.05 \times 10^{-34} \text{ Js}}{(9.11 \times 10^{-31} \text{ kg})(0.53 \times 10^{-10} \text{ m})}$$

$$= 2.17 \times 10^6 \text{ m/s}$$

$$\frac{\hbar}{L_1} = \frac{6.626 \times 10^{-34} \text{ Js}}{1.05 \times 10^{-34} \text{ kg m}^2/\text{s}} = 6.31 \text{ (close to } 2\pi)$$

$$\therefore \hbar = \frac{h}{2\pi}$$

⑥

$$r_2 = 4 (0.53 \times 10^{-10}) \text{ m}$$

$$V_2 = \frac{2 (1.05 \times 10^{-34} \text{ Js})}{4 (9.11 \times 10^{-31} \text{ kg})(0.53 \times 10^{-10} \text{ m})}$$

$$L_2 = 9.11 \times 10^{-31} \text{ kg} \times \frac{1}{2} (2.17 \times 10^6 \text{ m/s}) \times 4 (0.53 \times 10^{-10} \text{ m}) = 12.62 \text{ (close to } 4\pi)$$

$$\therefore L = nh, \quad \hbar = \frac{h}{2\pi}$$

d) $E = \frac{hc}{\lambda}$ $E_n = \frac{E_0}{n^2}$

$$E = \frac{6.6261 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ m/s}}{4.10 \times 10^{-9} \text{ m}} = 4.84 \times 10^{-19} \text{ J} = 3.026 \text{ eV}$$

$$E = E_{n_{\text{final}}} - E_{n_{\text{initial}}}$$

$$= E_0 \left(\frac{1}{(n_{\text{final}})^2} - \frac{1}{(n_{\text{initial}})^2} \right)$$

$$3.026 \text{ eV} = -13.6 \text{ eV} \left(\frac{1}{(n_{\text{final}})^2} - \frac{1}{(n_{\text{initial}})^2} \right)$$

$$-0.2225 = \frac{1}{(n_{\text{final}})^2} - \frac{1}{(n_{\text{initial}})^2} \quad \text{testing and plugging numbers}$$

we can find that $n_{\text{final}} = 6$, $n_{\text{initial}} = 2$

3a) $K = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$ $T = \frac{1}{1 + \frac{mV_0^2 a^2}{2E\hbar^2}}$

$$0.5 = \frac{1}{1 + \frac{9.1 \times 10^{-31} (25) (6.5 \times 10^{-9})^2}{2E (1.054 \times 10^{-35})^2}} = \frac{1}{1 + \frac{56.875 \times 10^{-49}}{E \cdot 2.22 \times 10^{-28}}}$$

$$0.5 = \frac{1}{1 + \frac{25.619 \times 10^{-21}}{E}} = \frac{E}{E + 25.619 \times 10^{-21}}$$

$$E = 50.619 \times 10^{-21} \text{ eV}$$

b) $\Psi(x) = a \cos(n\pi x/L) + b \sin(n\pi x/L)$ $P = \int_0^a |\Psi|^2 dx = 1$

$$|\Psi(x)|^2 = a^2 \cos^2(n\pi x/L) + 2ab \cos(n\pi x/L) \sin(n\pi x/L) + b^2 \sin^2(n\pi x/L)$$

$$\int_0^L a^2 \cos^2(n\pi x/L) dx = a^2 \left(L \sin\left(\frac{2n\pi x}{L}\right) + 2n\pi x \right) \Big|_0^L$$

$$= a^2 \frac{L}{2}$$

$$\int_0^L b^2 \sin^2(n\pi x/L) dx = -b^2 \left(L \sin\left(\frac{2n\pi x}{L}\right) - \frac{2n\pi x}{4\pi n} \right) \Bigg|_0^L$$

$$= -b^2 \left(-\frac{L}{2} \right)$$

$$\therefore a^2 \left(\frac{L}{2} \right) - b^2 \left(\frac{L}{2} \right) = 1$$

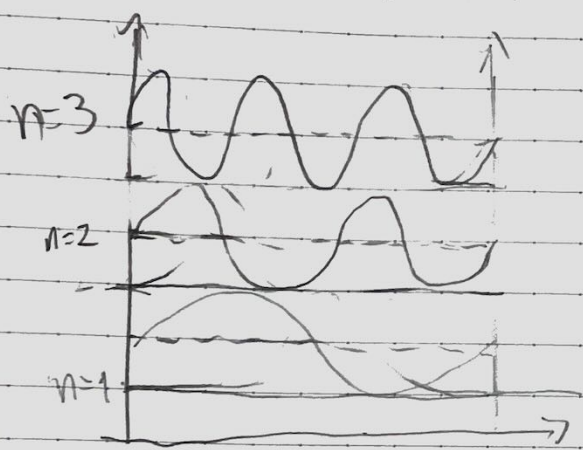
$$a^2 - b^2 = \frac{2}{L}$$

ii) $\Psi(x) = a \cos(n\pi x/L) + b \sin(n\pi x/L)$

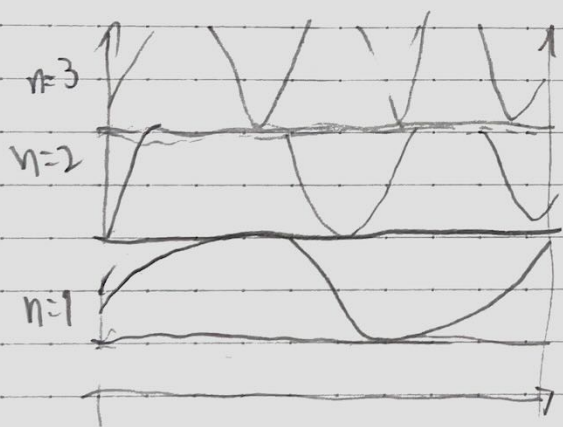
$$n=1: \Psi(x) = a \cos(\pi x/L) + b \sin(\pi x/L)$$

$$n=2: \Psi(x) = a \cos(2\pi x/L) + b \sin(2\pi x/L)$$

$$n=3: \Psi(x) = a \cos(3\pi x/L) + b \sin(3\pi x/L)$$



$$\Psi^2(x) = a^2 \cos^2(n\pi x/L) + 2ab \cos(n\pi x/L) \sin(n\pi x/L) + b^2 \sin^2(n\pi x/L)$$



(iii)

$$E_n = \frac{n^2 h^2}{8ma^2}$$

$$E_n = \frac{n^2 (6.63 \times 10^{-34} \text{ Js})^2}{8 (9.11 \times 10^{-31} \text{ kg}) (15 \times 10^{-9})^2}$$

$$E_1 = 3.524 \times 10^{-20} \text{ J}$$

$$E_2 = 14.097 \times 10^{-20} \text{ J}$$

$$E_3 = 31.7265 \times 10^{-20} \text{ J}$$

$$3.5 \text{ kJ/mol} < E_n < 8 \text{ kJ/mol}$$

Knowing that in 1 mol contains 6.022×10^{23} electrons

$$E_1 = 0.021 \text{ kJ/mol} \quad (\text{not in range})$$

$$E_2 = 0.849 \text{ kJ/mol} \quad (\text{in range})$$

$$E_3 = 1.929 \text{ kJ/mol} \quad (\text{in range})$$

$$\frac{3.5 \text{ kJ/mol}}{6.022 \times 10^{23}} = 5.82 \times 10^{-21} \text{ J/electron}, \quad \frac{8 \text{ kJ/mol}}{6.022 \times 10^{23}} = 1.33 \times 10^{-20} \text{ J/electron}$$

$$n_1^2 = \frac{8ma^2 E_n}{h^2} = \frac{8(9.11 \times 10^{-31} \text{ kg})(15 \times 10^{-9} \text{ m})^2 (5.82 \times 10^{-21} \text{ J/electron})}{(6.63 \times 10^{-34} \text{ Js})^2}$$

$$n_1^2 = 2.88$$

$$n_2^2 = \frac{8(9.11 \times 10^{-31} \text{ kg})(15 \times 10^{-9} \text{ m})^2 (1.33 \times 10^{-20} \text{ J/electron})}{(6.63 \times 10^{-34} \text{ Js})^2}$$

$$n_2^2 = 16.56$$

$16 - 2 = 14 \therefore$ There are 14 energy levels //

$$c) \quad \frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V) \psi = 0 \quad V = \frac{1}{2} kx^2 = 2\pi^2 m v^2 x^2$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} [E - 2\pi^2 m v^2 x^2] \psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \left[\frac{2mE}{\hbar^2} - \frac{4\pi^2 m^2 v^2 x^2}{\hbar^2} \right] \psi = 0 \quad \text{Wave equation}$$

$$a_{n+2} = \frac{2n+1-\lambda}{(n+2)(n+1)} a_n \quad \text{recursion formula}$$

$$2n+1-\lambda=0$$

$$\lambda = \frac{2E}{\hbar(2\pi\nu)}$$

$$\lambda = 2n+1$$

$$E = \left[n + \frac{1}{2}\right] \hbar \underbrace{(2\pi\nu)}_w \quad \therefore E_n = \left[n + \frac{1}{2}\right] \hbar w //$$

$$4a) \textcircled{i} - \frac{\sin^2 \theta}{r} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) - \frac{2\mu}{\hbar^2} r^2 \sin^2 \theta (E - V) - \frac{\sin \theta}{r} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right)$$

$$= \frac{1}{g} \frac{\partial^2 g}{\partial \phi^2}$$

$g = \psi$, $r = a$, $\mu = m$ and since it is moving through xy plane, $\theta = 90^\circ$

$$\therefore - \frac{2ma^2}{\hbar^2} (E - V) = \frac{1}{g} \frac{\partial^2 g}{\partial \phi^2}$$

$$V = -2T \text{ (inverse square)}$$

$$E = T + V \Rightarrow V = 2E$$

$$\frac{1}{a^2} \frac{d^2 \psi}{d\phi^2} + \frac{2m}{\hbar^2} |E| \psi = 0$$

$\textcircled{ii}$ This is similar to Bohr's assumptions because they both propose a circular motion for the electron

$\textcircled{iii}$ $\psi_{nlm}(r, \theta, \phi) = R_{nl}(r) Y_{lm}(\theta, \phi)$

$$L = \sqrt{l(l+1)}$$

$$E_n = -\frac{\mu}{2} \left(\frac{e^2}{4\pi\epsilon_0 \hbar} \right)^2 \frac{1}{n^2} = -\frac{E_0}{n^2}$$

$$b) R_{nl}(r) = - \left[\frac{(n-l-1)!}{2n[(n+l)!]^3} \right]^{\frac{1}{2}} \left(\frac{Z}{na_0} \right)^{l+\frac{3}{2}} r^l L_{n-l-1}^{2l+1} \left(\frac{2Zr}{na_0} \right) e^{-\frac{r}{na_0}}$$

$$R_{21} = \frac{r}{a_0} \frac{e^{-\frac{r}{2a_0}}}{\sqrt{3} (2a_0)^{\frac{3}{2}}}$$

$$E_n = -\frac{13.6 \text{ eV}}{n^2}$$

$$E_2 = -3.4 \text{ eV}$$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2\mu}{\hbar^2} \left[E - V - \frac{\hbar^2}{2\mu} \frac{l(l+1)}{r^2} \right] R = 0$$

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) + \frac{2\mu}{\hbar^2} \left[E - V - \frac{\hbar^2}{2\mu} \frac{2}{r^2} \right] R = 0$$

$$\frac{1}{\sin\theta} \frac{d}{d\theta} \left(\sin\theta \frac{dP}{d\theta} \right) + \left[2 - \frac{m_2^2}{\sin^2\theta} \right] P = 0$$

$$\frac{1}{R} \frac{\partial}{\partial r} \left(r^2 \frac{\partial R}{\partial r} \right) + \frac{2\mu r^2}{\hbar^2} (E - V) = \frac{m_2^2}{\sin\theta} - \frac{1}{r \sin\theta} \frac{\partial}{\partial \theta} \left(\sin\theta \frac{\partial P}{\partial \theta} \right)$$

It is consistent with Bohr model. //

$$c) R_{21} = \frac{r}{a_0} \frac{e^{-\frac{r}{2a_0}}}{\sqrt{3} (2a_0)^{\frac{3}{2}}}$$

$$R_{21}^2 = \frac{r^2 e^{-\frac{r}{a_0}}}{24a_0^5}$$

$$\int_0^\infty R_{21}^2 r^2 dr = 1$$

$$= \frac{1}{24a_0^5} \int_0^\infty r^4 e^{-\frac{r}{a_0}} dr$$

$$\int f g' = f g - \int f' g$$

$$u = -\frac{r}{a_0}$$

$$= \frac{1}{24a_0^5} \cdot a_0^5 \int_0^\infty u^4 e^u du$$

$$= \frac{1}{24a_0^5} \left[-a_0 (r^4 + 4a_0 r^3 + 12a_0^2 r^2 + 24a_0^3 r + 24a_0^4) e^{-\frac{r}{a_0}} \right]_0^\infty$$

$$= \frac{1}{24a_0^5} \cdot 24a_0^5 = 1 //$$